

Individual Assignment 5

Minh Tri Ngo

1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

Question 1

A: Planning

- x axis: Site

- y axis: log transformed Ostracod abundance
- geometry abundance: geom_jitter()
- geometry median: stat_summary()

B: Wrangle Data

```
#new object from aq_ins_clean
ostracod <- aq_ins_clean |>
  # filter to only include ostracod
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
slice_sample(ostracod, n=10)
```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2022-08-12_NVBR	2022-08-12 00:00:00	NVBR	ostr~	0	NA	NA	NA
2	2023-05-16_NVBR	2023-05-16 00:00:00	NVBR	ostr~	0.133	9.29	NA	9.83
3	2023-11-12_NMC	2023-11-12 00:00:00	NMC	ostr~	0.533	NA	47	14.0
4	2021-08-25_NPB	2021-08-25 00:00:00	NPB	ostr~	581.	7.52	4.57	4.5
5	2023-02-28_NPB1	2023-02-28 00:00:00	NPB1	ostr~	0	NA	NA	NA
6	2022-05-13_NVBR	2022-05-13 00:00:00	NVBR	ostr~	0	9.72	74.2	11.4
7	2022-01-24_NEC	2022-01-24 00:00:00	NEC	ostr~	0.267	NA	NA	NA
8	2023-05-05_NPB	2023-05-05 00:00:00	NPB	ostr~	24.7	NA	NA	12.1
9	2022-02-25_NEC	2022-02-25 00:00:00	NEC	ostr~	0	8.75	44.1	9.87
10	2022-01-28_NDC	2022-01-28 00:00:00	NDC	ostr~	9.47	7.65	2.00	9.55

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
comments <chr>, site_full <fct>, log_ave_lit <dbl>